Qwen_experiment Report

1.Executive Summary

This report presents a systematic empirical evaluation of the mathematical reasoning performance of Qwen2.5-Math-1.5B, a lightweight open-source large language model (LLM) specialized for mathematical tasks. The experiment was conducted on two standard mathematical reasoning benchmarks: the GSM8K test set (elementary arithmetic word problems) and the MATH-500 dataset (challenging competition-style math problems). Three widely adopted prompting strategies were compared: Chain-of-Thought (CoT), Least-to-Most (LTM), and Self-Consistency (SC, k=5).

Two core metrics were measured for all experiments:
1. **Accuracy:** The proportion of questions answered correctly, reflecting the model's reasoning correctness.
2. **Average Response Length**: The mean character count of model outputs, reflecting the verbosity and detail of reasoning processes.

Key findings from the 500-sample experiments on each dataset are as follows:

1. Accuracy Performance: Self-Consistency (k=5) consistently achieves the highest accuracy across both datasets, outperforming CoT and LTM by a clear margin. On GSM8K, SC reaches 61.8% accuracy, while CoT and LTM achieve 54.8% and 45.8% respectively. On MATH-500, SC scores 27.6%, compared to 26% for CoT and 21% for LTM.
2. Response Length Trade-off: SC delivers accuracy gains at the cost of significantly longer outputs, with average lengths exceeding 10,000 characters on both datasets (5x longer than CoT/LTM). LTM produces slightly longer responses than CoT but underperforms CoT in accuracy on both benchmarks.
3. Dataset Gap: The model shows a substantial performance gap between the two datasets, with GSM8K accuracy (54.8% for CoT) more than double that of MATH-500 (26% for CoT), highlighting the challenge of advanced mathematical reasoning for small LLMs.

2. Experimental Setup

2.1 Model

- Model: Qwen2.5-Math-1.5B
- Parameter Size: 1.5 Billion

- Model Type: Causal language model fine-tuned explicitly for mathematical reasoning tasks
- Precision: FP16 (half-precision floating point)
- Deployment: Single NVIDIA GPU with CUDA acceleration, optimized with torch.compile() for inference speedup

2.2 Datasets

Dataset	**Sample Size**	**Task Type**	**Difficulty**
GSM8K (Test Set)	500	Elementary arithmetic word problems	Low (K-12 grade school level)
MATH-500	500	Competition-style math problems (algebra, number theory, calculus, combinatorics)	High (Olympiad/AP level)

2.3 Prompting Methods

1. Chain-of-Thought (CoT): The standard step-by-step reasoning prompt, which guides the model to generate explicit intermediate reasoning steps before outputting the final answer.
2. Least-to-Most (LTM): A two-stage prompting method that first decomposes complex problems into smaller, solvable sub-problems, then solves them sequentially.
3. Self-Consistency (SC, k=5): A sampling-based method that generates 5 independent reasoning paths for each question, then selects the most frequent final answer as the output.

2.4 Evaluation Metrics

- Accuracy: Calculated as (Number of Correct Answers) / (Total Number of Questions), reported as a percentage.
- Average Response Length: Calculated as the mean character count of all model outputs for a given dataset and prompt method, reflecting the computational cost and reasoning verbosity.

3. Experimental Results

3.1 Full Result Summary

Dataset	Prompt Method	Total Samples	Correct	Accuracy	Avg Response Length
GSM8K	CoT	500	274	54.8%	1939.51
GSM8K	Least-to-Most	500	229	45.8%	2234.96
GSM8K	Self-Consistency (k=5)	500	309	61.8%	10227.44
MATH-500	CoT	500	130	26.0%	1910.05
MATH-500	Least-to-Most	500	105	21.0%	2114.98
MATH-500	Self-Consistency (k=5)	500	138	27.6%	10152.52

3.2 Key Result Analysis

1. Accuracy Ranking Across Methods:

- On GSM8K: Self-Consistency (61.8%) > CoT (54.8%) > Least-to-Most (45.8%)
- On MATH-500: Self-Consistency (27.6%) > CoT (26.0%) > Least-to-Most (21.0%)Self-Consistency is the top-performing method on both datasets, delivering a 7% absolute accuracy gain over CoT on GSM8K and a 1.6% gain on MATH-500. Notably, Least-to-Most underperforms CoT on both benchmarks, despite generating longer reasoning outputs.

2. Response Length Analysis:

- Self-Consistency produces ~5x longer responses than CoT and LTM on both datasets, with average lengths exceeding 10,000 characters. This is expected, as SC generates 5 independent reasoning paths per question.
- Least-to-Most generates slightly longer outputs than CoT (15% longer on GSM8K, 11% longer on MATH-500) but fails to translate this extra length into improved accuracy.
- CoT achieves the best balance of accuracy and efficiency, with moderate response lengths (~1900 characters) and strong performance relative to LTM.

3. Dataset Performance Gap:

- The model's accuracy on GSM8K is more than double its accuracy on

MATH-500 for all prompting methods, highlighting the significant challenge of advanced mathematical reasoning for small LLMs.

- The relative performance of prompting methods is consistent across datasets: SC > CoT > LTM, indicating that the effectiveness of prompting strategies is generalizable across different difficulty levels.
- The accuracy gap between SC and CoT is much larger on GSM8K (7%) than on MATH-500 (1.6%), suggesting that SC is more effective for simple, multi-step arithmetic problems than for complex competition math.

4. Discussion

The experimental results reveal critical insights into the behavior of small math-specialized LLMs under different prompting strategies:

1. Self-Consistency: Accuracy at the Cost of Efficiency: SC's superior accuracy comes with a 5x increase in inference cost (due to longer outputs and multiple sampling runs), making it suitable for scenarios where correctness is prioritized over speed. For resource-constrained deployments, CoT offers a much better efficiency-accuracy trade-off.
2. Least-to-Most Underperformance: Contrary to common findings in large LLMs, LTM fails to outperform CoT in this small model. This suggests that LTM's effectiveness relies on the model's inherent ability to decompose complex problems, a capability that small LLMs may lack. The longer outputs from LTM likely introduce redundant reasoning steps that harm final accuracy.
3. Dataset Difficulty and Prompting Efficacy: The model's performance gap between GSM8K and MATH-500 confirms that small LLMs struggle with abstract, high-level mathematical reasoning. The diminishing returns of SC on MATH-500 indicate that sampling multiple paths is less effective for solving complex problems that require deep domain knowledge, compared to simple arithmetic.
4. Response Length as a Proxy for Reasoning Effort: The results show that longer responses do not guarantee higher accuracy. LTM's longer outputs do not improve performance, while SC's extreme length delivers only marginal gains on hard problems. This highlights the importance of optimizing reasoning efficiency, not just length.